AI Messaging Sales Assistant

Planning document: WhatsApp, Instagram and Facebook Messenger, with n8n automation

Version 1.2 | Adds admin panel, Google Calendar and SaleSmartly reference | Status: plan for approval, no code built yet

1. Goal

Answer every customer message on WhatsApp, Instagram and Facebook Messenger within seconds, 24 hours a day, in Chinese and English. The assistant captures leads, books appointments, and hands the conversation to a staff member whenever a human is needed. It must be easy for three groups of people:

Who	What easy means for them
Customer	Instant reply in the chat app they already use, in their own language. A person is always one message away.
Staff	One inbox for all three channels, an AI summary on every handover, and a one-word command to give the chat back to the bot.
Business owner	A password-protected admin panel to change prices, FAQs and AI behaviour without a developer, a daily summary on the phone, and leads and bookings in one place.

2. Decisions already made

Topic	Decision
Scope	One business (not a multi-client product).
Industry (demo)	Aesthetic / beauty clinic. Swappable from the admin panel.
Languages	Chinese (Simplified and Traditional) and English.
First channel	WhatsApp. Instagram and Messenger follow using the same workflows.
Hosting	Self-hosted with **no paid hosting** (see section 4).
Automation	n8n (self-hosted, free community edition).
Unified inbox	Chatwoot (open source, self-hosted).
Admin system	Custom admin panel (web page, password-protected) inside the AI service.
Calendar	Google Calendar, two-way, **one calendar per practitioner**.

3. Recommended architecture

Four free open-source parts plus one AI service. Chatwoot gives staff the inbox and the handover feature, so none of that has to be built.

Layer	Tool	Role
Channels	Meta WhatsApp Cloud API, Instagram and Messenger APIs	Customer messages arrive from all three through one Meta app.
Inbox	Chatwoot (self-hosted)	Unified inbox, staff reply screen and mobile app, labels, assignment, conversation status.
Automation	n8n (self-hosted)	Routes messages, calls the AI, sends replies, runs handover alerts, reminders and the daily report.
AI service	Small Python service (FastAPI) in the aichat repo	Holds the prompt, the clinic knowledge, and the tools: availability, booking, lead save, handover.

Layer	Tool	Role
AI model	Claude via the Anthropic API	Understands Chinese and English, follows the rules, decides which tool to use.
Admin panel	Web pages in the same AI service	Owner and staff manage settings, services, FAQ, leads and bookings, and test the bot.
Calendar	Google Calendar API (free)	Source of busy time and home of every booking, one calendar per practitioner.
Database	PostgreSQL	Settings, conversations, leads, bookings, reminders. Also used by Chatwoot and n8n.

Message flow

```
Customer (WhatsApp / Instagram / Messenger)
        |
        v
   Meta webhook ---> Chatwoot (inbox)
                        |  webhook
                        v
               n8n: Inbound Router
                  |- staff already handling?  -> stay silent
                  |- call AI service /respond
                  |      |- tool: check_availability (Google Calendar)
                  |      |- tool: create_booking (writes calendar event)
                  |      |- tool: save_lead (score hot/warm/cold)
                  |      '- tool: handover_to_human
                  |- send reply through Chatwoot -> customer (target: a few seconds)
                  '- if handover: private note + open conversation + alert staff
```

4. Free hosting plan

Everything runs in Docker Compose on one machine. The only public address needed is Chatwoot, because Meta must reach its webhook. n8n and the AI service stay private.

Option	Cost	Good for	Watch out for
Always-free cloud VM (for example Oracle Cloud Always Free)	$0	True 24/7 running. Best fit for the goal.	Needs a card for identity check. Free capacity varies by region. Setup takes longer.
Your own PC or spare machine at the office	$0 (electricity)	Testing and pilot.	Must stay on 24/7. Needs 8 GB RAM recommended (Chatwoot is heavy), 4 GB minimum.
Public address: ngrok free static domain	$0	Stable webhook URL that does not change.	One free domain per account. Free-tier limits apply.
Public address: Cloudflare quick tunnel	$0	No account needed.	URL changes on every restart, so Meta webhook must be re-entered. Not for production.

Honest limits of free: a laptop that sleeps cannot answer messages at 3 a.m. For real 24/7 operation use an always-on machine. Also, the AI model is the one component that is not free to run (see section 11), but it is very cheap at this volume.

5. n8n workflows

#	Workflow	Trigger	What it does
1	Inbound router	Chatwoot webhook (new message)	Ignores staff and private notes, calls the AI service, posts the reply, runs handover steps. A private note with the text /bot from staff gives the chat back to the bot.
2	Booking reminders	Every 15 minutes	Asks the AI service which bookings are due, sends a reminder 24 hours and 2 hours before. Uses a WhatsApp template if the 24-hour window is closed.
3	Daily report	Every day, 9:00	Sends a summary to the owner on Telegram: new chats, leads by score, bookings, handovers, tomorrow's appointments.
4 (later)	Lead follow-up	Scheduled	Nudges hot leads who did not book, inside the allowed message window.

5a. Admin panel

A small password-protected web page, served by the AI service on the private network and reached through a secure link. Everything the owner would otherwise ask a developer to change is here. Changes apply on the next customer message, with no restart.

Page	What the owner or staff can do
Dashboard	Chats today, leads by hot / warm / cold, bookings, handovers waiting, bot status.
AI settings	Bot on or off (all channels or per channel), greeting, tone, handover rules, languages, after-hours message.
Services and prices	Add, edit or hide treatments with English and Chinese names, duration, price, and which practitioners can perform each one.
Practitioners	Add or remove practitioners, working days and hours, linked Google Calendar.
Hours and closures	Clinic opening hours and holiday dates.
Knowledge / FAQ	Edit bilingual answers, policies and promotions in a simple editor.
Leads	List, filter, mark status, export to CSV.
Bookings	View, cancel, reschedule, mark attended or no-show. Changes sync to Google Calendar.
Test chat	Talk to the bot in English or Chinese using the saved settings before going live.
Staff alerts	Telegram or email contacts for handovers and the daily report.
Users	Owner and staff logins with roles. Staff can manage bookings and leads, only the owner can change AI settings.

5b. Google Calendar, one per practitioner

The assistant never double-books and never offers a time a practitioner has blocked.

Part	How it works
Availability	For a service, the bot looks at the calendars of the practitioners who can do it, combines clinic hours, treatment length and existing events, and offers the real open slots. If a practitioner blocks time in their own calendar, it disappears from the offer.
Booking	The booking is saved and a calendar event is created on the chosen practitioner's calendar with customer name, phone, service and channel. Rechecked just before saving so two customers cannot take the same slot.
Change or cancel	Cancelling or rescheduling from chat or the admin panel updates or removes the event.
Practitioner choice	Customer can ask for a specific person. Otherwise the bot picks any available one.

Part	How it works
Connection	A Google service account. Each practitioner shares their calendar with it once. The Calendar API is free and needs no billing.
If Google is unreachable	The bot tells the customer it will confirm the time shortly, saves a pending request, and alerts staff. No false confirmations.

6. How the AI assistant behaves

Language

- Replies in the language of the customer's latest message, including mixed Chinese and English.
- Matches Simplified or Traditional Chinese to what the customer writes.
- Uses fixed treatment names from the business file, so names and prices never drift.

What it can do (tools)

Tool	Purpose
check_availability	Real open time slots for a service on a date, from Google Calendar free time of the practitioners who can do it, clinic hours and treatment length.
create_booking	Re-checks the slot, saves the appointment and creates the calendar event. Asks for name and phone first.
list_my_bookings / cancel_booking	Lets the customer see or cancel their own booking.
save_lead	Stores name, phone, interest, preferred time, language and a hot / warm / cold score.
handover_to_human	Marks the chat as staff-owned and passes a short summary.

Guardrails

- Answers only from the clinic knowledge. Never invents prices, promotions, or availability.
- No medical diagnosis, no guaranteed results. Medical questions go to staff.
- Tells the customer on the first message that it is a virtual assistant.
- Treats customer text as untrusted. Cannot be talked into changing prices or revealing its rules.
- If the AI service fails, the customer gets a polite message and staff are alerted, never silence.

Handover to a person happens when

- The customer asks for a human.
- Complaint, anger, refund or payment problem.
- Medical concern, allergy, side effect, pregnancy or similar.
- The question is outside the knowledge base, or the AI is unsure.
- The customer sends an image or voice note the assistant cannot read and it matters.

On handover n8n adds a private note with the AI summary, sets the conversation to Open, adds the label needs-human, and sends a Telegram alert. The bot stays silent until staff type /bot.

7. Demo use case: aesthetic clinic

Typical customer questions the assistant handles: treatment list and price ranges, how long a treatment takes, downtime and aftercare, who it is suitable for, location and parking, opening hours, current promotions, first-visit steps, and booking, rescheduling or cancelling.

Business content (services, prices, practitioners, hours, FAQ) is entered in the admin panel and stored in the database. The kit ships with bilingual sample content, loaded on first start.

All sample content will be clearly marked as sample and must be replaced with the real clinic's information and reviewed by the clinic before going live.

8. Meta platform rules to plan for

- **WhatsApp:** needs a Meta Business account and a phone number that is not already used on the normal WhatsApp app. Free-form replies are allowed only within 24 hours of the customer's last message. Outside that window only pre-approved templates can be sent.
- **Templates:** create English and Chinese versions of the appointment reminder and get them approved.
- **Instagram and Messenger:** the Meta app needs permissions approved through app review. This can take days to weeks, so start early.
- **Public HTTPS address** for the Chatwoot webhook (section 4).
- **Privacy:** short consent line when collecting name and phone, and an easy way to opt out.
- Check Meta's current pricing and rules before launch, because they change.

9. Data stored

Table	Holds
settings / services / practitioners	Everything edited in the admin panel: AI settings, FAQ text, treatments, practitioners and their calendar IDs, hours.
admin_users	Panel logins (hashed passwords) and roles.
conversations	Contact name, phone, channel, language, bot or human mode, handover reason and summary, time of last customer message.
messages	Recent chat history used as AI memory. Duplicate webhook deliveries are ignored.
leads	Name, phone, interest, preferred time, score, status, notes.
bookings	Customer, service, practitioner, Google event ID, start and end time, status, language, reminder flags.

Backups: a scheduled database dump kept off the machine (for example a free cloud drive). Secrets stay in a .env file that is never committed.

10. Build phases

Phase	Deliverable	Rough time
0. Accounts	Meta Business verification, WhatsApp number, free VM or PC, ngrok domain, Telegram bot for alerts, Anthropic API key	about 1 week, mostly waiting
1. Stack	Docker Compose: Chatwoot, n8n, Postgres, AI service, tunnel	2-3 days
2. Core bot	Bilingual answers from the knowledge files, WhatsApp reply loop	3-4 days
3. Leads and handover	Lead scoring, private note, staff alert, give-back command	2-3 days
4. Booking	Booking, cancel, reminders with templates	3-4 days
5. Google Calendar	Service account, one calendar per practitioner, availability and event sync	3-4 days
6. Admin panel	Login, dashboard, AI settings, services, practitioners, FAQ, leads, bookings, test chat	5-7 days
7. Channels	Turn on Instagram and Messenger	1-2 days plus Meta review
8. Polish	Daily report, backups, health alerts, real clinic content, staff training	3-5 days

Phase	Deliverable	Rough time
9. Optional extras	Staff translation view, customer tags and tiers, click-to-chat link and QR code, reactivation messages (see section 13)	as needed

Total about 5-6 weeks of work. Meta app review is the main schedule risk.

11. Running costs

Item	Cost	Note
Server / VM	$0	Always-free VM or an existing machine.
Google Calendar API	$0	Free for this volume. No billing account needed.
Chatwoot, n8n, Postgres, tunnel	$0	Open source or free tiers.
WhatsApp Cloud API	Low	Customer-started chats are inside the free service window; template messages (for example reminders outside the window) are charged by Meta. Confirm current rates.
Claude API (Haiku 5.5)	Pay per use, small	List price $0.10 per million input tokens and $0.50 per million output tokens. A typical chat is estimated at well under one US cent, so roughly a few dollars per 1,000 chats. Prompt caching lowers it further. This is the one unavoidable cost, because there is no free hosted model of this quality.

The model is set by one setting, so it can be raised to a stronger model later if answer quality needs it.

12. Risks and how we reduce them

Risk	Mitigation
Machine goes offline	Use an always-on VM. Health check alerts to Telegram.
Tunnel URL changes	Use ngrok static domain, not the quick tunnel.
AI gives a wrong price or claim	Answer only from the knowledge base, strict rules, staff review the sample content, review early chats.
Customer stuck with bot	Always-available human option, automatic handover rules.
Bot and staff both reply	Per-conversation mode: bot is silent whenever staff own the chat.
Meta approval delays	Start verification and app review in week 1. Launch WhatsApp first.
Google calendar not shared or token revoked	Admin dashboard shows calendar connection status per practitioner. Alert to staff if a check fails.
Admin panel exposed to the internet	Not published through the public tunnel. Strong passwords, roles, access over a private secure link.
Free tier limits change	Everything is plain Docker, so it can move to any other machine.

13. Reference: SaleSmartly

SaleSmartly (salesmartly.com) is a ready-made cloud platform for a very similar job: one inbox for many chat channels, AI agents with human takeover, a customer database, and broadcasts. Its app is behind a login, so I could not view its screens. This comparison uses its public website and one third-party review. Vendor claims are unverified, and no screen layouts were copied.

SaleSmartly module	What it offers (public information)	In our plan
Multi-channel inbox	Lists 14 channels, including WhatsApp, Messenger, Instagram, Telegram, LINE, WeChat, email and website chat. Keeps conversation history. Real-time translation.	Chatwoot inbox. WhatsApp, Instagram and Messenger first. Chatwoot also has website chat, Telegram, LINE and email. WeChat is not included.
AI agents + human	AI first reply, remembers customer needs, judges intent, follows up proactively, hands key chats to a person.	Same idea: AI with per-customer memory, lead scoring, handover rules. Proactive follow-up is workflow 4.
SCRM (customer records)	Company-owned customer data, tags and tiers, automatic assignment, routing links and QR codes, analytics.	Leads and bookings in the admin panel. Add tags and tiers, auto-assign handovers to staff, and a click-to-chat WhatsApp link and QR code.
Re-marketing	Targeted WhatsApp bulk messages by tag or intent, reactivation, automation.	Optional later: n8n campaign using approved templates, only to customers who opted in.
Attribution	Links chats to ad campaigns, Meta Conversion API.	Optional later.

Build with our plan or use SaleSmartly?

	Our plan (self-hosted)	SaleSmartly (cloud service)
Running cost	$0 hosting, plus small AI usage cost.	Monthly subscription. One third-party review quotes about $15.90 per month for a basic plan (2 agents, 10 accounts) and mentions a free plan. Check current prices.
Time to launch	About 5-6 weeks of build.	Days, because it is already built.
Data and control	Everything on your own server. Rules written exactly for your clinic.	Data held by the vendor. Customising limited to what the plan offers.
Booking with one Google Calendar per practitioner	Built to your rules, with calendar check before every booking.	Not confirmed in what I could read. Ask the vendor before relying on it.
Upkeep	You (or I) maintain it.	Vendor maintains it.
Both need	A verified Meta Business account and a WhatsApp number. Meta fees apply either way.	

Ideas we adopt: staff translation view (read Chinese chats in English and the reverse), customer tags and tiers, automatic assignment of handovers, a click-to-chat link and QR code for posters and the website, and customer memory. If speed matters more than control, trying SaleSmartly's free plan in parallel is a reasonable way to compare.

Sources: https://www.salesmartly.com/ and https://gptbots.ai/blog/salemartly-review

14. What I need from you to start building

- Approval of this plan, or changes you want.
- Confirm the demo industry stays as the aesthetic clinic, or give your own company's industry and services.
- Where it will run: always-free cloud VM (recommended) or an office machine.
- Time zone and currency for the business (the sample assumes Malaysia, ringgit).
- List of practitioners (names, what each can do, working hours) and a Google account that owns their calendars.
- When ready: Meta Business account, a WhatsApp number, an Anthropic API key, and a Telegram bot for alerts.

On approval, the next deliverable is the full starter kit in the aichat repository: Docker Compose stack, database schema, AI service with admin panel and Google Calendar sync, importable n8n workflows, bilingual sample

content, and a step-by-step setup guide.